



TECNOLÓGICO
NACIONAL DE MÉXICO



Tecnológico Nacional de México

Campus Saltillo

Ingeniería en Sistemas Computacionales

Computo en la nube

4.3 - Seguridad (hardening) en servidor web Apache 2

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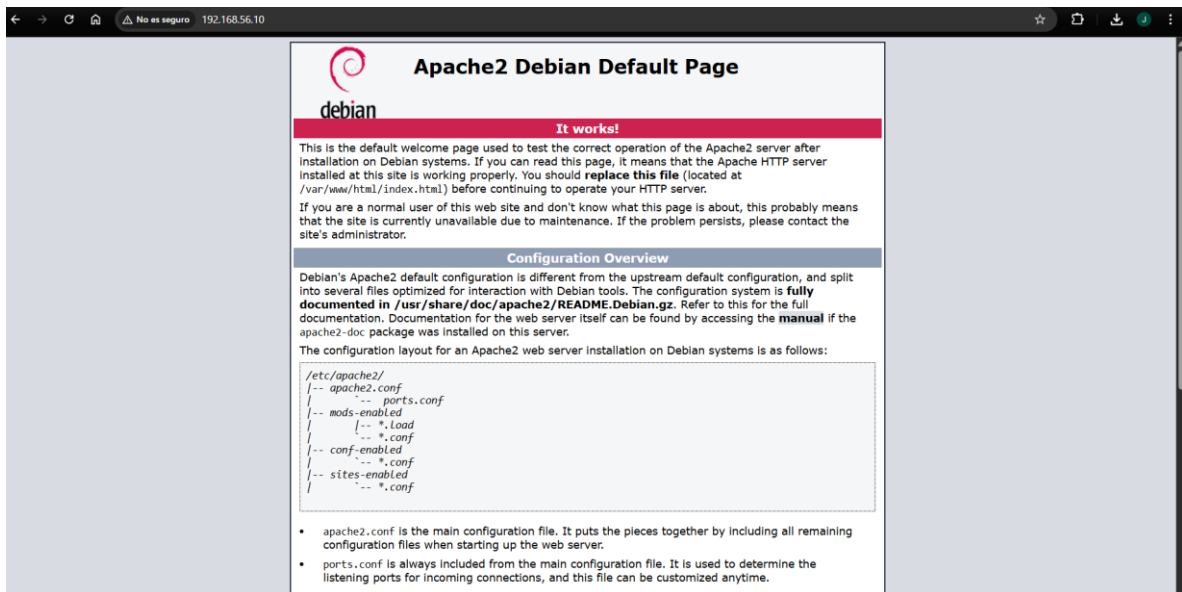
20/03/2026

1- Apache corriendo

```
jesusc@SRVREP:~$ sudo systemctl status apache2
sudo: unable to resolve host SRVAPP: Name or service not known
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-02-13 08:00:19 CST; 3h 21min ago
     Invocation: ec09df2e7e7247a0baca4d14f7270609
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 1498 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Process: 1716 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
   Main PID: 1502 (apache2)
      Tasks: 7 (limit: 2996)
     Memory: 14.1M (peak: 37.5M)
        CPU: 3.471s
    CGroup: /system.slice/apache2.service
            └─1502 /usr/sbin/apache2 -k start
              └─1725 /usr/sbin/apache2 -k start
                └─1726 /usr/sbin/apache2 -k start
                  └─1727 /usr/sbin/apache2 -k start
                    └─1728 /usr/sbin/apache2 -k start
                      └─1729 /usr/sbin/apache2 -k start
                        └─1806 /usr/sbin/apache2 -k start

Feb 13 08:00:19 SRVREP systemd[1]: Starting apache2.service - The Apache HTTP Server...
Feb 13 08:00:19 SRVREP apachectl[1501]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please see the documentation on how to set the servername directive, see: http://httpd.apache.org/docs/2.4/mod/mod\_main.html#servername
Feb 13 08:00:19 SRVREP systemd[1]: Started apache2.service - The Apache HTTP Server.
Feb 13 08:35:31 SRVREP systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Feb 13 08:35:32 SRVREP apachectl[1719]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please see the documentation on how to set the servername directive, see: http://httpd.apache.org/docs/2.4/mod/mod\_main.html#servername
Feb 13 08:35:32 SRVREP systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
```

2- Verificar funcionamiento en el navegador



3- Ocultar información sensible editando **sudo nano /etc/apache2/conf-available/security.conf**

```

#
# ServerTokens
# This directive configures what you return as the Server HTTP response
# Header. The default is 'Full' which sends information about the OS-Type
# and compiled in modules.
# Set to one of: Full | OS | Minimal | Minor | Major | Prod
# where Full conveys the most information, and Prod the least.
#ServerTokens Minimal
ServerTokens Prod
#ServerTokens Full

#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
# Set to one of: On | Off | EMail
#ServerSignature Off
ServerSignature off

#
# Allow TRACE method
#
# Set to "extended" to also reflect the request body (only for testing and
# diagnostic purposes).
#
# Set to one of: On | Off | extended
TraceEnable Off
#TraceEnable On

#
# Forbid access to version control directories
#
# If you use version control systems in your document root, you should
# probably deny access to their directories.

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line

```

4- Activar los cambios

```

jesusc@SRVREP:~$ sudo a2enconf security
sudo: unable to resolve host SRVAPP: Name or service not known
Conf security already enabled
jesusc@SRVREP:~$ |

```

5- Listar módulos de apache y desactivar algunos

```
jesusc@SRVAPP:~$ sudo a2dismod autoindex status proxy proxy_http
WARNING: The following essential module will be disabled.
This might result in unexpected behavior and should NOT be done
unless you know exactly what you are doing!
 autoindex

To continue type in the phrase 'Yes, do as I say!' or retry by passing '-f': Yes, d
o as I say!
Module autoindex disabled.
Module status disabled.
Module proxy already disabled
Module proxy_http already disabled
To activate the new configuration, you need to run:
  systemctl restart apache2
jesusc@SRVAPP:~$ sudo systemctl restart apache2
jesusc@SRVAPP:~$
```

6- Activar HTTPS (SSL)

[illegible]

```
# /usr/share/doc/apache2/README.Debian.gz for more info.
# If both key and certificate are stored in the same file,
# SSLCertificateFile directive is needed.
SSLCertificateFile /etc/apache2/ssl/apache.crt
SSLCertificateKeyFile /etc/apache2/ssl/apache.key
# Server Certificate Chain:
# Point SSLCertificateChainFile at a file containing the
```

← → ↻ 🏠 No es seguro https://192.168.56.10 ☆ 📁 ⬇️ J Nuevo Chrome disponible



Apache2 Debian Default Page



It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is **fully documented in /usr/share/doc/apache2/README.Debian.gz**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Debian systems is as follows:

```
/etc/apache2/
|-- apache2.conf
```

7- Configurar firewall

```
jesusc@SRVAPP:~$ sudo ufw allow 80
sudo ufw allow 443
sudo ufw enable
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
jesusc@SRVAPP:~$ |
```

8- Ver logs

```
jesusc@SRVAPP:~$ sudo tail -f /var/log/apache2/error.log
[Fri Feb 13 11:34:23.880750 2026] [core:notice] [pid 2336:tid 2336] AH00094: Command line: '/usr/sbin/apache2'
[Fri Feb 13 11:35:01.687981 2026] [mpm_prefork:notice] [pid 2336:tid 2336] AH00170: caught SIGWINCH, shutting down gracefully
[Fri Feb 13 11:35:02.181515 2026] [mpm_prefork:notice] [pid 2376:tid 2376] AH00163: Apache/2.4.66 (Debian) configured -- resuming normal operations
[Fri Feb 13 11:35:02.182193 2026] [core:notice] [pid 2376:tid 2376] AH00094: Command line: '/usr/sbin/apache2'
[Fri Feb 13 11:39:05.765610 2026] [mpm_prefork:notice] [pid 2376:tid 2376] AH00171: Graceful restart requested, doing restart
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
[Fri Feb 13 11:39:05.958200 2026] [ssl:warn] [pid 2376:tid 2376] AH01906: 127.0.1.1:443:0 server certificate is a CA certificate (BasicConstraints: CA == TRUE !?)
[Fri Feb 13 11:39:05.958267 2026] [ssl:warn] [pid 2376:tid 2376] AH01909: 127.0.1.1:443:0 server certificate does NOT include an ID which matches the server name
[Fri Feb 13 11:39:05.960962 2026] [mpm_prefork:notice] [pid 2376:tid 2376] AH00163: Apache/2.4.66 (Debian) OpenSSL/3.5.4 configured -- resuming normal operations
[Fri Feb 13 11:39:05.960997 2026] [core:notice] [pid 2376:tid 2376] AH00094: Command line: '/usr/sbin/apache2'
```

```
jesusc@SRVAPP:~$ sudo tail -f /var/log/apache2/access.log
192.168.56.1 - - [13/Feb/2026:09:03:16 -0600] "GET /usuarios/controladores/controlador.php?c=usuario&gui=usuario HTTP/1.1" 200 1365 "http://192.168.56.10/usuarios/controladores/controlador.php?c=usuario&a=Guardar&gui=usuario" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
192.168.56.1 - - [13/Feb/2026:09:12:51 -0600] "GET /usuarios/controladores/controlador.php?c=usuario&gui=usuario HTTP/1.1" 200 1429 "http://192.168.56.10/usuarios/controladores/controlador.php?c=usuario&a=Guardar&gui=usuario" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
192.168.56.1 - - [13/Feb/2026:09:12:52 -0600] "GET /favicon.ico HTTP/1.1" 404 531 "http://192.168.56.10/usuarios/controladores/controlador.php?c=usuario&gui=usuario" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
192.168.56.1 - - [13/Feb/2026:11:23:31 -0600] "-" 408 0 "-" "-"
192.168.56.1 - - [13/Feb/2026:11:23:31 -0600] "-" 408 0 "-" "-"
192.168.56.1 - - [13/Feb/2026:11:27:59 -0600] "GET / HTTP/1.1" 200 3367 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36"
192.168.56.1 - - [13/Feb/2026:11:39:10 -0600] "GET / HTTP/1.1" 200 3367 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36"
192.168.56.1 - - [13/Feb/2026:11:39:21 -0600] "GET / HTTP/1.1" 200 3367 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36"
192.168.56.1 - - [13/Feb/2026:11:39:22 -0600] "GET / HTTP/1.1" 200 3366 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36"
192.168.56.1 - - [13/Feb/2026:11:39:23 -0600] "GET / HTTP/1.1" 200 3366 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36"
```

^C

```
jesusc@SRVAPP:~$ |
```

Conclusión

En esta práctica se logró implementar el hardening de Apache, reduciendo la superficie de ataque mediante la desactivación de módulos innecesarios, ocultando información sensible y habilitando HTTPS. Esto me permitió comprender la importancia de asegurar un servidor web para evitar vulnerabilidades y accesos no autorizados.